

Step 1: Explore the Table

Check table structure

```
1 PRAGMA table_info(superstore);
```

	cid	name	type	notnull	dfit_value	pk
1	0	Row ID	INTEGER	0	NULL	0
2	1	Order ID	TEXT	0	NULL	0
3	2	Order Date	TEXT	0	NULL	0
4	3	Ship Date	TEXT	0	NULL	0
5	4	Ship Mode	TEXT	0	NULL	0
6	5	Customer ID	TEXT	0	NULL	0
7	6	Customer Name	TEXT	0	NULL	0
8	7	Segment	TEXT	0	NULL	0
9	8	Country	TEXT	0	NULL	0
10	9	City	TEXT	0	NULL	0
11	10	State	TEXT	0	NULL	0
12	11	Postal Code	INTEGER	0	NULL	0
13	12	Region	TEXT	0	NULL	0
14	13	Product ID	TEXT	0	NULL	0
15	14	Category	TEXT	0	NULL	0
16	15	Sub-Category	TEXT	0	NULL	0
17	16	Product Name	TEXT	0	NULL	0
18	17	Sales	REAL	0	NULL	0
19	18	Quantity	INTEGER	0	NULL	0
20	19	Discount	INTEGER	0	NULL	0
21	20	Profit	REAL	0	NULL	0

View first 10 rows

```
1 SELECT *
2 FROM superstore
3 LIMIT 10;
```

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcase
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances

Step 2: Validate Data

Total rows

11	--Step 2: Validate Data
12	--Total rows
13	SELECT COUNT(*) AS total_rows
14	FROM superstore;
15	
total_rows	
1	9994

Step 3: Apply WHERE Filters

Filter by Region

SQL 1*

Step 4: Aggregations using GROUP BY

Total Sales by Region

```
45 --Step 4: Aggregations using GROUP BY
46 --Total Sales by Region
47 SELECT Region,
48        SUM(Sales) AS Total_Sales
49 FROM superstore
50 GROUP BY Region;
51
```

	Region	Total_Sales
1	Central	501239.8908
2	East	678781.24
3	South	391721.905
4	West	725457.8245

Total Quantity by Category

```
74
75 --Top Categories by Sales
76 SELECT Category,
77        SUM(Sales) AS Total_Sales
78 FROM superstore
79 GROUP BY Category
80 ORDER BY Total_Sales DESC;
81
82
```

	Category	Total_Sales
1	Technology	836154.033
2	Furniture	741999.7953
3	Office Supplies	719047.032

Step 5: Sort and Limit Results

Top Categories by Sales

```
74
75 --Top Categories by Sales
76 SELECT Category,
77        SUM(Sales) AS Total_Sales
78 FROM superstore
79 GROUP BY Category
80 ORDER BY Total_Sales DESC;
81
82
```

	Category	Total_Sales
1	Technology	836154.033
2	Furniture	741999.7953
3	Office Supplies	719047.032

Step 6: Business Use Cases

Monthly Sales Trend

```
83 --Step 6: Business Use Cases
84 --Monthly Sales Trend
85 SELECT strftime('%Y-%m', "Order Date") AS Month,
86        SUM(Sales) AS Total_Sales
87 FROM superstore
88 GROUP BY Month
89 ORDER BY Month;
90
```

	Region	Total_Records
1	Central	2323
2	East	2848
3	South	1620
4	West	3203

Top 10 Customers

```
90
91 --Top 10 Customers
92 SELECT "Customer Name",
93        SUM(Sales) AS Total_Sales
94 FROM superstore
95 GROUP BY "Customer Name"
96 ORDER BY Total_Sales DESC
97 LIMIT 10;
98
```

	Customer Name	Total_Sales
1	Sean Miller	25043.05
2	Tamara Chand	19052.218
3	Raymond Buch	15117.339
4	Tom Ashbrook	14595.62
5	Adrian Barton	14473.571
6	Ken Lonsdale	14175.229
7	Sanjit Chand	14142.334
8	Hunter Lopez	12873.298
9	Sanjit Engle	12209.438
10	Christopher Conant	12129.072

Step 7: Final Validation

Count Records by Region

```
107 --Step 7: Final Validation
108 --Count Records by Region
109 SELECT Region,
110        COUNT(*) AS Total_Records
111 FROM superstore
112 GROUP BY Region;
113
```

	Region	Total_Records
1	Central	2323
2	East	2848
3	South	1620
4	West	3203